

Lebesgue Outer Measure: Definition and Translation Invariance

Understanding measure and its
invariance under translation

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Definition of Lebesgue Outer Measure

1. **Definition :** For any set A of real numbers, consider a sequence of non empty open bounded

intervals $\{I_k\}_{k=1}^{\infty}$ such that $A \subseteq \bigcup_{k=1}^{\infty} I_k$. We define Lebesgue outer measure of A by,

$$m^*(A) = \inf \left\{ \sum_{k=1}^{\infty} \ell(I_k) \mid A \subseteq \bigcup_{k=1}^{\infty} I_k \right\}$$

Proposition : Outer measure is translation invariant. i.e. for any set A and for any real number y.

$$m^*(A+y) = m^*(A).$$

Proof : Let A be any subset of $\mathbb{R}$. If there is a countable collection of open intervals $\{I_k\}_{k=1}^{\infty}$ such that

$$A \subseteq \bigcup_{k=1}^{\infty} I_k, \text{ then}$$

$$A \subseteq \bigcup_{k=1}^{\infty} I_k \Leftrightarrow A + y \subseteq \left(\bigcup_{k=1}^{\infty} I_k \right) + y$$

$$\Leftrightarrow A + y \subseteq \bigcup_{k=1}^{\infty} (I_k + y)$$

Also $\ell(I_k) = \ell(I_k + y)$ for all $k = 1, 2, 3, \dots$

$$\begin{aligned}
\text{Therefore, } m^*(A) &= \inf \left\{ \sum_{k=1}^{\infty} \ell(I_k) \mid A \subseteq \bigcup_{k=1}^{\infty} I_k \right\} \\
&= \inf \left\{ \sum_{k=1}^{\infty} \ell(I_k) \mid A + y \subseteq \bigcup_{k=1}^{\infty} (I_k + y) \right\} \\
&= \inf \left\{ \sum_{k=1}^{\infty} \ell(I_k + y) \mid A + y \subseteq \bigcup_{k=1}^{\infty} (I_k + y) \right\} \\
&= \inf \left\{ \sum_{k=1}^{\infty} \ell(J_k) \mid A + y \subseteq \bigcup_{k=1}^{\infty} J_k \right\} \\
&= m^*(A + y)
\end{aligned}$$

SIMPLE TWO-STEP EXAMPLE

Translation Invariance: Basic Interval Shift

Evaluating the outer measure of interval $A = [1, 4]$ with translation $y = 5$

STEP 01

Measure of Set A

$$\begin{aligned}\text{Interval: } A &= [1, 4] \\ m^*(A) &= 4 - 1 = 3\end{aligned}$$

Length of segment $[1, 4]$ equals 3.

STEP 02

Shift & Measure of Set $A + 5$

$$\begin{aligned}A + 5 &= [1 + 5, 4 + 5] = [6, 9] \\ m^*(A + 5) &= 9 - 6 = 3\end{aligned}$$

Length of shifted interval remains 3.

Conclusion: $m^*(A + 5) = m^*(A) = 3$

POSITION SHIFTS, SIZE STAYS CONSTANT

Thank You